1)

Scenario 1: Π_1 and Π_2 are coincident (so they are the same plane)
Scenario 2: Π_1 and Π_2 are parallel
Scenario 3: Π_1 and Π_2 intersect at a line

2)

Scenario 1: Π_1, Π_2 and Π_3 intersect at a point (one solution)
Scenario 2: Π_1, Π_2 and Π_3 are coincident (infinitely many solutions)
Scenario 3: Π_1 and Π_2 are coincident, while Π_3 is parallel (no solutions)
Scenario 4: Π_1 and Π_2 are coincident, while Π_3 intersects at a line (infinitely many solutions)
Scenario 5: Π_1, Π_2 and Π_3 are parallel (no solutions)
Scenario 6: Π_1 and Π_2 are parallel, while Π_3 intersects Π_1 and Π_2 at different lines (no solutions)
Scenario 7: Π_1, Π_2 and Π_3 form a triangular prism (no solutions)
Scenario 8: Π_1, Π_2 and Π_3 form a sheaf (so they all intersect at a line, and consequently there are infinitely many solutions)

3)

Scenario 1: Π_1, Π_2 and Π_3 intersect at a point (one solution)
Scenario 2: Π_1, Π_2 and Π_3 form a triangular prism (no solutions)
Scenario 3: Π_1, Π_2 and Π_3 form a sheaf (so they all intersect at a line, and consequently there are infinitely many solutions)

4)

Π_1, Π_2 and Π_3 form a sheaf (so they all intersect at a line, and consequently there are infinitely many solutions)

5) a)

$$\mathbf{A}^{-1} = \begin{pmatrix} -8 & 13 & -6 \\ \frac{11}{2} & -\frac{17}{2} & 4 \\ \frac{13}{2} & -\frac{21}{2} & 5 \end{pmatrix}$$

5) b)

$$\begin{pmatrix}1 & 4 & -2\\ 3 & 2 & 2\\ 5 & -1 & 7\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}10\\ 12\\ 8\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1 & 4 & -2\\ 3 & 2 & 2\\ 5 & -1 & 7\end{pmatrix}^{-1}\begin{pmatrix}10\\ 12\\ 8\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}-8 & 13 & -6\\ \frac{11}{2} & -\frac{17}{2} & 4\\ \frac{13}{2} & -\frac{21}{2} & 5\end{pmatrix}\begin{pmatrix}10\\ 12\\ 8\end{pmatrix}=\begin{pmatrix}28\\ -15\\ -21\end{pmatrix}$$

So $x=28$, $y=-15$, $z=-21$

5) c)

The three planes intersect at a unique point, $(28,-15,-21)$

6)

$$\begin{pmatrix}2 & 3 & -1\\ 1 & 2 & 1\\ 4 & -1 & -1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}2 & 3 & -1\\ 1 & 2 & 1\\ 4 & -1 & -1\end{pmatrix}^{-1}\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}-\frac{1}{22} & \frac{2}{11} & \frac{5}{22}\\ \frac{5}{22} & \frac{1}{11} & -\frac{3}{22}\\ -\frac{9}{22} & \frac{7}{11} & \frac{1}{22}\end{pmatrix}\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}=\begin{pmatrix}1\\ 0\\ 1\end{pmatrix}$$

So $x=1$, $y=0$, $z=1$

The three planes intersect at a unique point, $(1,0,1)$

7)

$$\begin{pmatrix}1 & 4 & -2\\ -6 & -7 & 1\\ 2 & 5 & -2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}15\\ -40\\ 22\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1 & 4 & -2\\ -6 & -7 & 1\\ 2 & 5 & -2\end{pmatrix}^{-1}\begin{pmatrix}15\\ -40\\ 22\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}9 & -2 & -10\\ -10 & 2 & 11\\ -16 & 3 & 17\end{pmatrix}\begin{pmatrix}15\\ -40\\ 22\end{pmatrix}=\begin{pmatrix}-5\\ 12\\ 14\end{pmatrix}$$

So $x=-5$, $y=12$, $z=14$

The three planes intersect at a unique point, $(-5,12,14)$

8)

$$\begin{vmatrix} 4 & 1 & 9 \\ 3 & 2 & -5 \\ 4 & 3 & -10 \end{vmatrix} = -1 \neq 0$$

As the determinant is nonzero, the three planes will always meet at a unique point, irrespective of the value of k

9) a)

Π_1 can be written as $3x + y - 2z = \frac{1}{2}$ so it is parallel to Π_3

Π_2 is not parallel to Π_1 or Π_3

Hence Π_1 and Π_3 are parallel, while Π_2 intersects Π_1 and Π_3 at different lines (no solutions)

9) b)

Π_2 can be written as $3x - y + 5z = 4$, which is the same equation as Π_3

Π_1 is not parallel to Π_2 or Π_3

Hence Π_2 and Π_3 are coincident, while Π_1 intersects at a line (infinitely many solutions)

9) c)

Π_1 can be written as $4x + 6y - 3z = -\frac{6}{5}$ (by multiplying the original equation through by $\frac{1}{5}$)

Π_2 can be written as $4x + 6y - 3z = -\frac{6}{5}$ (by multiplying the original equation through by 3)

Π_3 can be written as $4x + 6y - 3z = -\frac{6}{5}$ (by multiplying the original equation through by $\frac{2}{7}$)

Hence Π_1, Π_2 and Π_3 are coincident (infinitely many solutions)

10)

None of the planes are parallel (or coincident)

$\begin{vmatrix} 11 & -3 & 1 \\ 1 & 0 & 2 \\ 3 & -1 & -1 \end{vmatrix} = 0$ so the planes do not intersect at a unique point.

Let $z = 0$:

$11x - 3y = 2$ (1)

$x = -5$ (2)

$3x - y = 4$ (3)

Solve (1) and (2): $x = -5$ and $y = -19$

Check in (3):

$3(-5) - (-19) = 4$

So the equations are consistent, so the planes must form a sheaf (infinitely many solutions).

11)

None of the planes are parallel (or coincident)

$\begin{vmatrix} 4 & -21 & 6 \\ 8 & 2 & 4 \\ 10 & 8 & 4 \end{vmatrix} = 0$ so the planes do not intersect at a unique point.

Let $z = 0$:

$4x - 21y = 10$ (1)

$8x + 2y = 3$ (2)

$10x + 8y = 7$ (3)

Solve (1) and (2): $x = \frac{83}{176}$ and $y = -\frac{17}{44}$

Check in (3):

$10\left(\frac{83}{176}\right) + 8\left(-\frac{17}{44}\right) = \frac{13}{8} \neq 7$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

12)

None of the planes are parallel (or coincident)

$\begin{vmatrix} 5 & 6 & 26 \\ 2 & -3 & -4 \\ 1 & -3 & -6 \end{vmatrix} = 0$ so the planes do not intersect at a unique point.

Let $z = 0$:

$5x + 6y = 6$ (1)

$2x - 3y = 6$ (2)

$x - 3y = 12$ (3)

Solve (1) and (2): $x = 2$ and $y = -\frac{2}{3}$

Check in (3):

$(2) - 3\left(-\frac{2}{3}\right) = 4 \neq 12$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

13) a)

None of the planes are parallel (or coincident)

When $k = 0$, then $x = 0$, $y = 0$, $z = 0$ is a solution. So all three planes pass through $(0, 0, 0)$. Therefore, the three planes intersect at a unique point, $(0, 0, 0)$.

13) b)

$\begin{vmatrix} 2 & 4 & 1 \\ 1 & 19 & -8 \\ -2 & 4 & -5 \end{vmatrix} = 0$ so the planes do not intersect at a unique point.

Let $z = 0$:

$2x + 4y = k$ (1)

$x + 19y = 0$ (2)

$-2x + 4y = 0$ (3)

Solve (2) and (3): $x = 0$ and $y = 0$

Check in (3):

$2(0) + 4(0) = 0 \neq k$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

14) a)

When $k = 1$:

$\Pi_1: 2x + y + z = 1$

$\Pi_2: x + 3y + z = 1$

$\Pi_3: 4x + 2y + 2z = 1$

Π_1 and Π_3 are parallel to one another, as Π_3 can be written as $2x + y + z = \frac{1}{2}$ (by multiplying through by $\frac{1}{2}$)

So we have that Π_1 and Π_3 are parallel, while Π_2 intersects Π_1 and Π_3 at different lines (no solutions)

14) b)

When $k = 3$:

$\Pi_1: 2x + 3y + 3z = 3$

$\Pi_2: 3x + 3y + 3z = 3$

$\Pi_3: 4x + 2y + 2z = 3$

None of the planes are parallel (or coincident)

$\begin{vmatrix} 2 & 3 & 3 \\ 3 & 3 & 3 \\ 4 & 2 & 2 \end{vmatrix} = 0$ so the planes do not intersect at a unique point.

Let $z = 0$:

$2x + 3y = 3$ (1)

$3x + 3y = 3$ (2)

$4x + 2y = 3$ (3)

Solve (1) and (2): $x = 0$ and $y = 1$

Check in (3):

$4(0) + 2(1) = 2 \neq 3$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

15) a)

As the planes do not intersect at a unique point, we must have that $\begin{vmatrix} 2 & 4 & 5 \\ 3 & k & 4 \\ -2 & 4 & k \end{vmatrix} = 0$

$\Rightarrow 2(k \times k - 4 \times 4) - 4\big(3 \times k - 4 \times (-2)\big) + 5\big(3 \times 4 - k \times (-2)\big) = 0$

$\Rightarrow 2(k^2 - 16) - 4(3k + 8) + 5(12 + 2k) = 0$

$\Rightarrow 2k^2 - 32 - 12k - 32 + 60 + 10k = 0$

$\Rightarrow 2k^2 - 2k - 4 = 0$

$\Rightarrow k^2 - k - 2 = 0$

$\Rightarrow (k - 2)(k + 1) = 0$

$\Rightarrow k = 2$ or $k = -1$

15) b)

When $k = 2$:

$\Pi_1: 2x + 4y + 5z = 23$

$\Pi_2: 3x + 2y + 4z = 20$

$\Pi_3: -2x + 4y + 2z = 6$

None of the planes are parallel (or coincident)

Let $z = 0$:

$2x + 4y = 23$ (1)

$3x + 2y = 20$ (2)

$-2x + 4y = 6$ (3)

Solve (1) and (2): $x = \frac{17}{4}$ and $y = \frac{29}{8}$

Check in (3):

$-2\left(\frac{17}{4}\right) + 4\left(\frac{29}{8}\right) = 6$

So the equations are consistent, so the planes must form a sheaf (infinitely many solutions).

When $k = -1$:

$\Pi_1: 2x + 4y + 5z = 23$

$\Pi_2: 3x - y + 4z = 20$

$\Pi_3: -2x + 4y - z = 6$

None of the planes are parallel (or coincident)

Let $z = 0$:

$2x + 4y = 23$ (1)

$3x - y = 20$ (2)

$-2x + 4y = 6$ (3)

Solve (1) and (2): $x = \frac{103}{14}$ and $y = \frac{29}{14}$

Check in (3):

$-2\left(\frac{103}{14}\right) + 4\left(\frac{29}{14}\right) = -\frac{45}{7} \neq 6$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

16)

$$\begin{vmatrix} k+1 & 3 & -2 \\ k & 4 & -\frac{1}{4}(k+8) \\ 0 & 4 & -1 \end{vmatrix} = (k+1)\left(4\times(-1)-\left(-\tfrac{1}{4}(k+8)\right)\times 4\right)-3\left(k\times(-1)+\left(-\tfrac{1}{4}(k+8)\right)\times 0\right)-2(k\times 4-4\times 0)$$

$$= (k+1)\left(-4+(k+8)\right)-3(-k)-2(4k)$$

$$= (k+1)(k+4)+3k-8k$$

$$= k^2+5k+4+3k-8k$$

$= k^2+4 \geq 4$ for all real values of k

So the planes always meet at a unique point.

17) a)

$$\begin{vmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{vmatrix} = k(3\times k-k\times 2)-0(1\times k-k\times 1)-1(1\times 2-3\times 1)$$

$$= k(k)-0-1(-1)$$

$= k^2+1 \geq 1$ for all real values of k

So the planes always meet at a unique point.

17) b)

$$\begin{pmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{pmatrix}^{-1}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Find the cofactor matrix:

$$\begin{pmatrix} k & 0 & -1 \\ -2 & k^2+1 & -2k \\ 3 & -k^2-1 & 3k \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} k & -2 & 3 \\ 0 & k^2+1 & -k^2-1 \\ -1 & -2k & 3k \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{vmatrix} = k^2+1$$

Then:

$$\begin{pmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{pmatrix}^{-1} = \frac{1}{k^2+1}\begin{pmatrix} k & -2 & 3 \\ 0 & k^2+1 & -k^2-1 \\ -1 & -2k & 3k \end{pmatrix} = \begin{pmatrix} \frac{k}{k^2+1} & -\frac{2}{k^2+1} & \frac{3}{k^2+1} \\ 0 & 1 & -1 \\ -\frac{1}{k^2+1} & -\frac{2k}{k^2+1} & \frac{3k}{k^2+1} \end{pmatrix}$$

So:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{k}{k^2+1} & -\frac{2}{k^2+1} & \frac{3}{k^2+1} \\ 0 & 1 & -1 \\ -\frac{1}{k^2+1} & -\frac{2k}{k^2+1} & \frac{3k}{k^2+1} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{2}{k^2+1} \\ 1 \\ -\frac{2k}{k^2+1} \end{pmatrix}$$

Therefore, the three planes intersect at a unique point, $\left(-\frac{2}{k^2+1}, 1, -\frac{2k}{k^2+1}\right)$.

18) a)

As the planes do not intersect at a unique point, we must have that

$$\begin{vmatrix} 1 & k & -4 \\ -6 & 4 & -3 \\ k & 1 & 4 \end{vmatrix} = 0$$

$\Rightarrow 1(4 \times 4 - (-3) \times 1) - k\big((-6) \times 4 - (-3) \times k\big) - 4\big((-6) \times 1 - 4 \times k\big) = 0$

$\Rightarrow 1(16 + 3) - k(-24 + 3k) - 4(-6 - 4k) = 0$

$\Rightarrow 19 + 24k - 3k^2 + 24 + 16k = 0$

$\Rightarrow -3k^2 + 40k + 43 = 0$

$\Rightarrow 3k^2 - 40k - 43 = 0$

$\Rightarrow (3k - 43)(k + 1) = 0$

$\Rightarrow k = \frac{43}{3}$ or $k = -1$

18) b)

When $k = \frac{43}{3}$:

$\Pi_1: x + \frac{43}{3}y - 4z = -19$

$\Pi_2: -6x + 4y - 3z = -42$

$\Pi_3: \frac{43}{3}x + y + 4z = 19$

None of the planes are parallel (or coincident)

Let $z = 0$:

$x + \frac{43}{3}y = -19$ (1)

$-6x + 4y = -42$ (2)

$\frac{43}{3}x + y = 19$ (3)

Solve (1) and (2): $x = \frac{263}{45}$ and $y = -\frac{26}{15}$

Check in (3):

$\frac{43}{3}\left(\frac{263}{45}\right) + \left(-\frac{26}{15}\right) = \frac{2215}{27} \neq 19$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

When $k = -1$:

$\Pi_1: x - y - 4z = -19$

$\Pi_2: -6x + 4y - 3z = -42$

$\Pi_3: -x + y + 4z = 19$

The equation for Π_3 can be written as $x - y - 4z = -19$ (by multiplying through by -1), and this is the same equation as Π_1

Therefore Π_1 and Π_3 are coincident, while Π_2 (which is not parallel to Π_1 or Π_3) intersects Π_1 and Π_3 at a line (infinitely many solutions)

19) a)

As the planes do not intersect at a unique point, we must have that $\begin{vmatrix} k & 2 & 3k \\ 4 & k & -6 \\ 1 & 0 & -6 \end{vmatrix} = 0$

$\Rightarrow k(k \times (-6) - (-6) \times 0) - 2(4 \times (-6) - (-6) \times 1) + 3k(4 \times 0 - k \times 1) = 0$

$\Rightarrow k(-6k) - 2(-24 + 6) + 3k(-k) = 0$

$\Rightarrow -6k^2 + 36 - 3k^2 = 0$

$\Rightarrow -9k^2 + 36 = 0$

$\Rightarrow k^2 - 4 = 0$

$\Rightarrow (k - 2)(k + 2) = 0$

$\Rightarrow k = 2$ or $k = -2$

19) b)

When $k = 2$:

$\Pi_1: 2x + 2y + 6z = 2$

$\Pi_2: 4x + 2y - 6z = -2$

$\Pi_3: x - 6z = 0$

None of the planes are parallel (or coincident)

Let $z = 0$:

$2x + 2y = 2$ (1)

$4x + 2y = -2$ (2)

$x = 0$ (3)

Solve (2) and (3): $x = 0$ and $y = -1$

Check in (1):

$2(0) + 2(-1) = -2 \neq 2$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

When $k = -2$:

$\Pi_1: -2x + 2y - 6z = 2$

$\Pi_2: 4x - 2y - 6z = -2$

$\Pi_3: x - 6z = 0$

None of the planes are parallel (or coincident)

Let $z = 0$:

$-2x + 2y = 2$ (1)

$4x - 2y = -2$ (2)

$x = 0$ (3)

Solve (2) and (3): $x = 0$ and $y = 1$

Check in (1):

$2(0) + 2(1) = 2$

So the equations are consistent, so the planes must form a sheaf (infinitely many solutions).

20) a)

As the planes do not intersect at a unique point, we must have that

$$\begin{vmatrix} -5 & 3 & k+3 \\ k & 2k & -4 \\ 2 & -9 & 1 \end{vmatrix} = 0$$

$\Rightarrow -5\big(2k \times 1 - (-4) \times (-9)\big) - 3(k \times 1 - (-4) \times 2) + (k+3)(k \times (-9) - 2k \times 2) = 0$

$\Rightarrow -5(2k - 36) - 3(k + 8) + (k + 3)(-13k) = 0$

$\Rightarrow -10k + 180 - 3k - 24 - 13k^2 - 39k = 0$

$\Rightarrow -13k^2 - 52k + 156 = 0$

$\Rightarrow k^2 + 4k - 12 = 0$

$\Rightarrow (k + 6)(k - 2) = 0$

$\Rightarrow k = -6$ or $k = 2$

20) b)

When $k = -6$:

$\Pi_1: -5x + 3y - 3z = 2$

$\Pi_2: -6x - 12y - 4z = -5$

$\Pi_3: 2x - 9y + z = 4$

None of the planes are parallel (or coincident)

Let $z = 0$:

$-5x + 3y = 2$ (1)

$-6x - 12y = -5$ (2)

$2x - 9y = 4$ (3)

Solve (1) and (2): $x = -\frac{3}{26}$ and $y = \frac{37}{78}$

Check in (3):

$2\left(-\frac{3}{26}\right) - 9\left(\frac{37}{78}\right) = -\frac{9}{2} \neq 4$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

When $k = 2$:

$\Pi_1: -5x + 3y + 5z = 2$

$\Pi_2: 2x + 4y - 4z = -5$

$\Pi_3: 2x - 9y + z = 4$

None of the planes are parallel (or coincident)

Let $z = 0$:

$-5x + 3y = 2$ (1)

$2x + 4y = -5$ (2)

$2x - 9y = 4$ (3)

Solve (1) and (2): $x = -\frac{23}{26}$ and $y = -\frac{21}{26}$

Check in (3):

$2\left(-\frac{23}{26}\right) - 9\left(-\frac{21}{26}\right) = \frac{11}{2} \neq 4$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

21) a)

As the planes do not intersect at a unique point, we must have that

$$\begin{vmatrix} 4 & -2 & 3 \\ k+1 & 3 & k-2 \\ 4 & k-3 & 5 \end{vmatrix} = 0$$

$\Rightarrow 4\big(3\times 5-(k-2)\times(k-3)\big)+2\big((k+1)\times 5-(k-2)\times 4\big)+3\big((k+1)\times(k-3)-3\times 4\big)=0$

$\Rightarrow 4\big(15-(k^2-5k+6)\big)+2(5k+5-4k+8)+3(k^2-2k-3-12)=0$

$\Rightarrow 4(9+5k-k^2)+2(k+13)+3(k^2-2k-15)=0$

$\Rightarrow 36+20k-4k^2+2k+26+3k^2-6k-45=0$

$\Rightarrow -k^2+16k+17=0$

$\Rightarrow k^2-16k-17=0$

$\Rightarrow (k-17)(k+1)=0$

$\Rightarrow k=17$ or $k=-1$

21) b)

When $k = 17$:

$\Pi_1: 4x - 2y + 3z = 17$

$\Pi_2: 18x + 3y + 15z = 15$

$\Pi_3: 4x + 14y + 5z = -65$

None of the planes are parallel (or coincident)

Let $z = 0$:

$4x - 2y = 17$ (1)

$18x + 3y = 15$ (2)

$4x + 14y = -65$ (3)

Solve (1) and (2): $x = \frac{27}{16}$ and $y = -\frac{41}{8}$

Check in (3):

$4\left(\frac{27}{16}\right) + 14\left(-\frac{41}{8}\right) = -65$

So the equations are consistent, so the planes must form a sheaf (infinitely many solutions).

When $k = -1$:

$\Pi_1: 4x - 2y + 3z = 17$

$\Pi_2: 3y - 3z = 15$

$\Pi_3: 4x - 4y + 5z = -65$

None of the planes are parallel (or coincident)

Let $z = 0$:

$4x - 2y = 17$ (1)

$3y = 15$ (2)

$4x - 4y = -65$ (3)

Solve (1) and (2): $x = \frac{27}{4}$ and $y = 5$

Check in (3):

$4\left(\frac{27}{4}\right) - 4(5) = 7 \neq -65$

So the equations are inconsistent, so the planes must form a triangular prism (no solutions).

22)

If the planes form a sheaf, then we must have that $\begin{vmatrix} 2k & k+3 & k \\ 1 & -1 & 2 \\ 3 & 1 & -2 \end{vmatrix} = 0$

$\Rightarrow 2k\big((-1)\times(-2)-2\times 1\big)-(k+3)(1\times(-2)-2\times 3)+k(1\times 1-(-1)\times 3)=0$

$\Rightarrow 2k(0)-(k+3)(-8)+k(4)=0$

$\Rightarrow 8k+24+4k=0$

$\Rightarrow 12k+24=0$

$\Rightarrow k=-2$

When $k=-2$:

$\Pi_1: -4x+y-2z=3$

$\Pi_2: x-y+2z=p$

$\Pi_3: 3x+y-2z=1$

None of the planes are parallel (or coincident)

For the planes to form a sheaf, the equations must be consistent.

Let $z=0$:

$-4x+y=3$ (1)

$x-y=p$ (2)

$3x+y=1$ (3)

Solve (1) and (3): $x=-\frac{2}{7}$ and $y=\frac{13}{7}$

Substitute into (2):

$\left(-\frac{2}{7}\right)-\left(\frac{13}{7}\right)=p \Rightarrow p=-\frac{15}{7}$

So in order for the planes to form a sheaf, we must have $k=-2$ and $p=-\frac{15}{7}$

23) a)

As the planes do not intersect at a unique point, we must have that

$\begin{vmatrix} 6 & -1 & k \\ 2 & -3 & 5 \\ 4 & 2 & -3 \end{vmatrix} = 0$

$\Rightarrow 6\big((-3)\times(-3)-5\times 2\big)+1(2\times(-3)-5\times 4)+k(2\times 2-(-3)\times 4)=0$

$\Rightarrow 6(-1)+(-26)+k(16)=0$

$\Rightarrow -6-26+16k=0$

$\Rightarrow 16k-32=0$

$\Rightarrow k=2$

23) b)

$m(6x-y)+n(2x-3y)\equiv 4x+2y$

$(6m+2n)x+(-m-3n)y\equiv 4x+2y$

Comparing coefficients:

$6m+2n=4$ (1)

$-m-3n=2$ (2)

Solve (1) and (2): $m=1$ and $n=-1$

23) c)

Let $z=0$:

$6x-y=33$

$2x-3y=25$

$4x+2y=p$

$1\times 33-1\times 25=p\Rightarrow p=8$

24) a)

$\mathbf{A}^{-1}$ cannot exist if $\det\mathbf{A}=0$

$\det\mathbf{A}=0$

$\Rightarrow 3\big(10\times 3-0\times(k-1)\big)+4(1\times 3-0\times 2)+17(1\times(k-1)-10\times 2)=0$

$\Rightarrow 3(30)+4(3)+17(k-21)=0$

$\Rightarrow 90+12+17k-357=0$

$\Rightarrow 17k-255=0$

$\Rightarrow k=15$

$\therefore \mathbf{A}^{-1}$ exists if $k\neq 15$

24) b)

i.e. $k=14$

$$\begin{pmatrix}3&-4&17\\1&10&0\\2&13&3\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}17\\0\\m\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}3&-4&17\\1&10&0\\2&13&3\end{pmatrix}^{-1}\begin{pmatrix}17\\0\\m\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{30}{17}&-\frac{233}{17}&10\\\frac{3}{17}&\frac{25}{17}&-1\\\frac{7}{17}&\frac{47}{17}&-2\end{pmatrix}\begin{pmatrix}17\\0\\m\end{pmatrix}=\begin{pmatrix}10m-30\\3-m\\7-2m\end{pmatrix}$$

So the planes intersect at $(10m-30,3-m,7-2m)$

24) c)

i.e. $k = 15$

None of the planes are parallel (or coincident)

Let $z = 0$:

$3x - 4y = n$ (1)

$x + 10y = 1$ (2)

$2x + 14y = 2$ (3)

Solve (2) and (3): $x = 1$ and $y = 0$

Substitute into (1):

$3(1) - 4(0) = n \Rightarrow n = 3$

As the planes do not meet at a unique point and the equations are consistent, we must have that the planes form a sheaf (infinitely many solutions).

25) a)

As the planes do not intersect at a unique point, we must have that

$$\begin{vmatrix} 2 & -3 & k \\ 5 & -3 & k^2 \\ 4 & 12 & -1 \end{vmatrix} = 0$$

$\Rightarrow 2\big((-3) \times (-1) - k^2 \times 12\big) + 3(5 \times (-1) - k^2 \times 4) + k(5 \times 12 - (-3) \times 4) = 0$

$\Rightarrow 2(3 - 12k^2) + 3(-5 - 4k^2) + k(72) = 0$

$\Rightarrow 6 - 24k^2 - 15 - 12k^2 + 72k = 0$

$\Rightarrow -36k^2 + 72k - 9 = 0$

$\Rightarrow 4k^2 - 8k + 1 = 0$

$\Rightarrow k = 1 \pm \frac{\sqrt{3}}{2}$

25) b)

When $k = 1 + \frac{\sqrt{3}}{2}$:

$\Pi_1: 2x - 3y + \left(1 + \frac{\sqrt{3}}{2}\right)z = p$

$\Pi_2: 5x - 3y + \left(\frac{7}{4} + \sqrt{3}\right)z = 1$

$\Pi_3: 4x + 12y - z = 8$

None of the planes are parallel (or coincident)

If the equations were consistent:

Let $z = 0$:

$2x - 3y = p$ (1)

$5x - 3y = 1$ (2)

$4x + 12y = 8$ (3)

Solve (2) and (3): $x = \frac{1}{2}$ and $y = \frac{1}{2}$

Substitute into (1):

$2\left(\frac{1}{2}\right) - 3\left(\frac{1}{2}\right) = p \Rightarrow p = -\frac{1}{2}$

When $k = 1 - \frac{\sqrt{3}}{2}$:

$\Pi_1: 2x - 3y + \left(1 - \frac{\sqrt{3}}{2}\right)z = p$

$\Pi_2: 5x - 3y + \left(\frac{7}{4} - \sqrt{3}\right)z = 1 - \sqrt{3}$

$\Pi_3: 4x + 12y - z = 8$

None of the planes are parallel (or coincident)

If the equations were consistent:

Let $z = 0$:

$2x - 3y = p$ (1)

$5x - 3y = 1 - \sqrt{3}$ (2)

$4x + 12y = 8$ (3)

Solve (2) and (3): $x = \frac{1}{2} - \frac{\sqrt{3}}{6}$ and $y = \frac{1}{2} + \frac{\sqrt{3}}{18}$

Substitute into (1):

$2\left(\frac{1}{2} - \frac{\sqrt{3}}{6}\right) - 3\left(\frac{1}{2} + \frac{\sqrt{3}}{18}\right) = p \Rightarrow p = -\frac{1}{2} - \frac{\sqrt{3}}{2}$

So in order for the equations to be inconsistent, $k = 1 + \frac{\sqrt{3}}{2}$ and $p \neq -\frac{1}{2}$,

or $k = 1 - \frac{\sqrt{3}}{2}$ and $p \neq -\frac{1}{2} - \frac{\sqrt{3}}{2}$

25) c)

As the planes do not meet at a unique point and the equations are inconsistent, we must have that the planes form a triangular prism (no solutions)

26) a)

$$\mathbf{AB} = \begin{pmatrix} k-1 & 2-k & 1 \\ k+2 & 1 & -3 \\ k+1 & 2 & -5 \end{pmatrix}\begin{pmatrix} 1 & -8 & k+1 \\ 15 & -20 & 15 \\ 7 & -16 & 15 \end{pmatrix}$$

$$= \begin{pmatrix} 36-14k & 12k-48 & k^2-15k+44 \\ k-4 & 12-8k & k^2+3k-28 \\ k-4 & 32-8k & k^2+2k-44 \end{pmatrix}$$

26) b)

When $k = 4$:

$$\mathbf{AB} = \begin{pmatrix} -20 & 0 & 0 \\ 0 & -20 & 0 \\ 0 & 0 & -20 \end{pmatrix}$$

$$\mathbf{A}\begin{pmatrix} 1 & -8 & 5 \\ 15 & -20 & 15 \\ 7 & -16 & 15 \end{pmatrix} = \begin{pmatrix} -20 & 0 & 0 \\ 0 & -20 & 0 \\ 0 & 0 & -20 \end{pmatrix}$$

$$\Rightarrow \mathbf{A}\begin{pmatrix} 1 & -8 & 5 \\ 15 & -20 & 15 \\ 7 & -16 & 15 \end{pmatrix} = -20\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \mathbf{A}\begin{pmatrix} -\frac{1}{20} & \frac{2}{5} & -\frac{1}{4} \\ -\frac{3}{4} & 1 & -\frac{3}{4} \\ -\frac{7}{20} & \frac{4}{5} & -\frac{3}{4} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore \mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{20} & \frac{2}{5} & -\frac{1}{4} \\ -\frac{3}{4} & 1 & -\frac{3}{4} \\ -\frac{7}{20} & \frac{4}{5} & -\frac{3}{4} \end{pmatrix}$$

26) c)

$$\mathbf{A}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}40\\100\\-1000\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\mathbf{A}^{-1}\begin{pmatrix}40\\100\\-1000\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{1}{20}&\frac{2}{5}&-\frac{1}{4}\\-\frac{3}{4}&1&-\frac{3}{4}\\-\frac{7}{20}&\frac{4}{5}&-\frac{3}{4}\end{pmatrix}\begin{pmatrix}40\\100\\-1000\end{pmatrix}=\begin{pmatrix}288\\820\\816\end{pmatrix}$$

So the planes intersect at the unique point $(288, 820, 816)$

27) a)

$\mathbf{A}$ is singular when $\det\mathbf{A}=0$

$$\begin{vmatrix}7&-1&2\\k&4&-2\\1&-6&3\end{vmatrix}=0$$

$$\Rightarrow 7\big(4\times3-(-2)\times(-6)\big)+1(k\times3-(-2)\times1)+2(k\times(-6)-4\times1)=0$$

$$\Rightarrow 7(0)+(3k+2)+2(-6k-4)=0$$

$$\Rightarrow 3k+2-12k-8=0$$

$$\Rightarrow -9k-6=0$$

$$\Rightarrow k=-\frac{2}{3}$$

27) b)

$$\begin{pmatrix} 7 & -1 & 2 \\ -\frac{7}{9} & 4 & -2 \\ 1 & -6 & 3 \end{pmatrix}\begin{pmatrix} \sqrt{x} \\ \sqrt{y} \\ \sqrt{z} \end{pmatrix} = \begin{pmatrix} 49 \\ \frac{10}{3} \\ -6 \end{pmatrix} \Rightarrow \begin{pmatrix} \sqrt{x} \\ \sqrt{y} \\ \sqrt{z} \end{pmatrix} = \begin{pmatrix} 7 & -1 & 2 \\ -\frac{7}{9} & 4 & -2 \\ 1 & -6 & 3 \end{pmatrix}^{-1}\begin{pmatrix} 49 \\ \frac{10}{3} \\ -6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \sqrt{x} \\ \sqrt{y} \\ \sqrt{z} \end{pmatrix} = \begin{pmatrix} 0 & -9 & -6 \\ \frac{1}{3} & 19 & \frac{112}{9} \\ \frac{2}{3} & 41 & \frac{245}{9} \end{pmatrix}\begin{pmatrix} 49 \\ \frac{10}{3} \\ -6 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \\ 6 \end{pmatrix}$$

$\sqrt{x} = 6 \Rightarrow x = 36$

$\sqrt{y} = 5 \Rightarrow y = 25$

$\sqrt{z} = 6 \Rightarrow z = 36$

$\therefore$ the three surfaces intersect at $(36, 25, 36)$

28) a)

Π_1 and Π_2 share a common point of intersection, namely $(1, 2, 5)$

Check whether Π_1 and Π_2 are coincident:

$$\begin{pmatrix}2\\1\\3\end{pmatrix} \times \begin{pmatrix}3\\-1\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 3\\3 & -1 & 2\end{vmatrix} = \begin{pmatrix}5\\5\\-5\end{pmatrix}$$

$$\begin{pmatrix}4\\-1\\2\end{pmatrix} \times \begin{pmatrix}3\\2\\1\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\4 & -1 & 2\\3 & 2 & 1\end{vmatrix} = \begin{pmatrix}-5\\2\\11\end{pmatrix}$$

Π_1 and Π_2 have different normal vectors, so they are not coincident.

Check whether $(1, 2, 5)$ lies on Π_3:

$1 = 8 - 2p - q$ (1)

$2 = -3 + 4p - q$ (2)

$5 = 2 - 3p + 3q$ (3)

Solve (1) and (2): $p = 2$ and $q = 3$
Check in (3):
$2 - 3(2) + 3(3) = 5$
So $(1, 2, 5)$ lies on Π_3

Hence the three planes must intersect at a unique point, $(1, 2, 5)$

28) b)

There are no obvious common points of intersection.

Consider the planes' normal vectors to determine if any are parallel (or potentially coincident):

$$\Pi_1: \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} \times \begin{pmatrix} 3 \\ 9 \\ 5 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 5 & -2 \\ 3 & 9 & 5 \end{vmatrix} = \begin{pmatrix} 43 \\ -11 \\ -6 \end{pmatrix}$$

$$\Pi_2: \begin{pmatrix} 6 \\ 18 \\ 10 \end{pmatrix} \times \begin{pmatrix} 3 \\ 15 \\ -6 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 18 & 10 \\ 3 & 15 & -6 \end{vmatrix} = \begin{pmatrix} -258 \\ 66 \\ 36 \end{pmatrix} = -6 \begin{pmatrix} 43 \\ -11 \\ -6 \end{pmatrix}$$

So Π_1 and Π_2 are at least parallel

$(3, 4, 1)$ lies on Π_1, so check if this point lies on Π_2:

$3 = 3 + 6m + 3n$ (1)

$4 = 10 + 18m + 15n$ (2)

$1 = -10 + 10m - 6n$ (3)

Solve (1) and (2): $m = \frac{1}{2}$ and $n = -1$

Check in (3):

$-10 + 10\left(\frac{1}{2}\right) - 6(-1) = 1$

So $(3, 4, 1)$ lies on Π_2, so Π_1 and Π_2 are coincident.

Now consider the normal vector of Π_3 to determine whether it is parallel (or potentially coincident), or if it intersects Π_1 and Π_2 at a line:

$$\Pi_3: \begin{pmatrix} 3 \\ 9 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 10 \\ -4 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 9 & 4 \\ 2 & 10 & -4 \end{vmatrix} = \begin{pmatrix} -76 \\ 20 \\ 12 \end{pmatrix}$$

This normal vector is not a multiple of the others, and so while Π_1 and Π_2 are coincident, Π_3 intersects at a line.